

# DNS

## Question 1

### Trame 1

| ETH                          | IP                     | UDP              | DNS           |
|------------------------------|------------------------|------------------|---------------|
| @Mac dst : d0:e0:42:80:14:55 | @IP dst :172.31.25.142 | port src : 54956 | flag : 0x0100 |
| @Mac src : cc:96:e5:1c:35:cd | @IP src:172.18.26.101  | port dst : 53    |               |

### Trame 2

| ETH                          | IP                      | UDP              | DNS           |
|------------------------------|-------------------------|------------------|---------------|
| @Mac dst : cc:96:e5:1c:35:cd | @IP dst : 172.18.26.101 | port src :53     | flag : 0x8580 |
| @Mac src : d0:e0:42:80:14:55 | @IP src : 172.31.25.142 | port dst : 54956 |               |

## Question 2

### Trame 1

| ETH                          | IP                     | UDP              | DNS           |
|------------------------------|------------------------|------------------|---------------|
| @Mac dst : d0:e0:42:80:14:55 | @IP dst :172.31.25.142 | port src : 59759 | flag : 0x0100 |
| @Mac src : cc:96:e5:1c:35:cd | @IP src:172.18.26.101  | port dst : 53    |               |

### Trame 2

| ETH                          | IP                      | UDP              | DNS           |
|------------------------------|-------------------------|------------------|---------------|
| @Mac dst : cc:96:e5:1c:35:cd | @IP dst : 172.18.26.101 | port src :53     | flag : 0x8580 |
| @Mac src : d0:e0:42:80:14:55 | @IP src : 172.31.25.142 | port dst : 59759 |               |

# DHCP

## Question 2

Ces échanges ont pour but d'attribuer automatiquement à la machine d'ip 0.0.0.0 une ip , un masque , une passerelle , un DNS.

## Question 3

### Trame 1 DHCP discover

| ETH                             | IP                           | UDP           | DHCP                                                             |
|---------------------------------|------------------------------|---------------|------------------------------------------------------------------|
| @MAC src :<br>00:0b:82:01:fc:42 | @IP src : 0.0.0.0            | Port src : 68 | Message type : Boot<br>request<br>DHCP message<br>type(Discover) |
| @MAC dst :<br>ff:ff:ff:ff:ff:ff | @IP dst :<br>255.255.255.255 | Port dst : 67 |                                                                  |

### Trame 2 DHCP Offer

| ETH                             | IP                        | UDP           | DHCP                                                                             |
|---------------------------------|---------------------------|---------------|----------------------------------------------------------------------------------|
| @MAC src :<br>00:08:74:ad:f1:9b | @IP src : 192.168.0.1     | Port src : 67 | Message type : Boot<br>reply(1)<br>Your (client) IP<br>address :<br>192.168.0.10 |
| @MAC dst :<br>00:0b:82:01:fc:42 | @IP dst :<br>192.168.0.10 | Port dst : 68 |                                                                                  |

#### Trame 3 DHCP request

| ETH                             | IP                           | UDP           | DHCP                                                                        |
|---------------------------------|------------------------------|---------------|-----------------------------------------------------------------------------|
| @MAC src :<br>00:0b:82:01:fc:42 | @IP src :0.0.0.0             | Port src : 68 | Message type : Boot<br>request<br>Requested IP<br>address<br>(192.168.0.10) |
| @MAC dst :<br>ff:ff:ff:ff:ff:ff | @IP dst :<br>255.255.255.255 | Port dst : 67 |                                                                             |

#### Trame 4 DHCP ACK

| ETH                             | IP                        | UDP           | DHCP                                                                                                                                                |
|---------------------------------|---------------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| @MAC src :<br>00:08:74:ad:f1:9b | @IP src :192.168.0.1      | Port src : 67 | Message type : Boot<br>reply<br>Your (client) IP<br>address :<br>192.168.0.10<br>DHCP Message type<br>(ACK)<br>Rebinding time value<br>: 52 min 30s |
| @MAC dst :<br>00:0b:82:01:fc:42 | @IP dst :<br>192.168.0.10 | Port dst : 68 |                                                                                                                                                     |

# SSH

#### **Question 2**

Cet échange a pour but de connecter notre machine à distance avec un équipement du réseau de l'ut de bethune grâce à un protocole sécurisé

#### **Question 3**

#### Trame 1(Protocol)

| ETH                             | IP                        | TCP                                          | SSH                                                                                                |
|---------------------------------|---------------------------|----------------------------------------------|----------------------------------------------------------------------------------------------------|
| @MAC src :<br>cc:96ce5:1c:35:cd | @IP src<br>:172.31.25.142 | Port src : 50564<br>Port dst : 22<br>ACK : 1 | Protocol :<br>SSH-2.0-OpenSSH_<br>9 6p1<br>Ubuntu-3ubuntu13.13<br>[Direction: client to<br>server] |
| @MAC dst :<br>2a:91:45:2f:a3:a9 | @IP dst : 172.31.25.9     |                                              |                                                                                                    |

#### Trame 2(Protocol)

| ETH                             | IP                         | TCP                                           | SSH                                                                                               |
|---------------------------------|----------------------------|-----------------------------------------------|---------------------------------------------------------------------------------------------------|
| @MAC src :<br>2a:91:45:2f:a3:a9 | @IP src :172.31.25.9       | Port src : 22<br>Port dst : 50564<br>ACK : 44 | Protocol :<br>SSH-2.0-OpenSSH_<br>8 9p1<br>Ubuntu-3ubuntu0.13<br>[Direction: server to<br>client] |
| @MAC dst :<br>cc:96ce5:1c:35:cd | @IP dst :<br>172.31.25.142 |                                               |                                                                                                   |

#### Trame 3(Key Exchange Init)

| ETH                             | IP                         | TCP                                           | SSH                                                                                                                                                      |
|---------------------------------|----------------------------|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| @MAC src :<br>2a:91:45:2f:a3:a9 | @IP src :172.31.25.9       | Port src : 22<br>Port dst : 50564<br>ACK : 44 | SSH Version 2<br>(encryption: <a href="#">chacha20</a><br><a href="#">-poly1305@openssh.com</a><br>compression:none)<br>[Direction: server to<br>client] |
| @MAC dst :<br>cc:96ce5:1c:35:cd | @IP dst :<br>172.31.25.142 |                                               |                                                                                                                                                          |

#### Trame 4 (Key Exchange init)

| ETH                             | IP                        | TCP                                             | SSH                                                                                                                                                       |
|---------------------------------|---------------------------|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| @MAC src :<br>cc:96ce5:1c:35:cd | @IP src<br>:172.31.25.142 | Port src : 50564<br>Port dst : 22<br>ACK : 1155 | SSH Version 2<br>(encryption: <a href="#">chacha20</a><br><a href="#">-poly1305@openssh.com</a><br>compression:none)<br>[Direction: client to<br>serveur] |
| @MAC dst :<br>2a:91:45:2f:a3:a9 | @IP dst : 172.31.25.9     |                                                 |                                                                                                                                                           |

#### Trame 5 (Diffie-Hellman Key Exchange Init)

| ETH                             | IP                         | TCP                                            | SSH                                                                                                                               |
|---------------------------------|----------------------------|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| @MAC src :<br>2a:91:45:2f:a3:a9 | @IP src :172.31.25.9       | Port src :22<br>Port dst : 50564<br>ACK : 2788 | SSH Version 2<br>(encryption: <a href="#">chacha20-poly1305@openssh.com</a><br>compression:none)<br>[Direction: server to client] |
| @MAC dst :<br>cc:96ce5:1c:35:cd | @IP dst :<br>172.31.25.142 |                                                |                                                                                                                                   |

#### Trame 6

| ETH                             | IP                     | TCP                                             | SSH                                                                                                                               |
|---------------------------------|------------------------|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| @MAC src :<br>2a:91:45:2f:a3:a9 | @IP src :172.31.25.142 | Port src : 50564<br>Port dst : 22<br>ACK : 2719 | SSH Version 2<br>(encryption: <a href="#">chacha20-poly1305@openssh.com</a><br>compression:none)<br>[Direction: client to server] |
| @MAC dst :<br>cc:96ce5:1c:35:cd | @IP dst : 172.31.25.9  |                                                 |                                                                                                                                   |

## FTP

#### Trame 1

| Trames | ETH                                                             | IP | TCP | FTP |
|--------|-----------------------------------------------------------------|----|-----|-----|
| 1      | @MAC src:<br>2a:91:45:2f:a3:a9<br><br>@MAC dst :<br>cc:96:e5:1c |    |     |     |
| 2      |                                                                 |    |     |     |
| 3      |                                                                 |    |     |     |
| 4      |                                                                 |    |     |     |
| 5      |                                                                 |    |     |     |
| 6      |                                                                 |    |     |     |
| 7      |                                                                 |    |     |     |
| 8      |                                                                 |    |     |     |

|    |  |  |  |  |
|----|--|--|--|--|
| 9  |  |  |  |  |
| 10 |  |  |  |  |
| 11 |  |  |  |  |
| 12 |  |  |  |  |
| 13 |  |  |  |  |
| 14 |  |  |  |  |
| 15 |  |  |  |  |
| 16 |  |  |  |  |
| 17 |  |  |  |  |
| 18 |  |  |  |  |